SPONSOR CHALLENGES AND PROJECT SELECTION

ShellHacks 2026 project recommendation

Eight confirmed challenges
Fourteen candidate concepts

Preferred direction: ReplayBench with Tiger Data. Prove that delayed telemetry produces correct summaries, then show its measured performance.

DECISION BASIS	STATUS
Challenge scope	Eight MLH challenges apply to 2026 by explicit user confirmation; additional challenges remain possible.
Team and time	Four teammates; skills and remaining time unspecified. Full 36-hour plan is a conditional baseline.
Deliverable	Research and project selection only. No project implementation or sponsor account setup performed.

Version 1.2.0 • Evidence reviewed 26 September 2026 • Independent analysis for the team
Updated companion to the 2017–2025 winner study; existing Smartsheet-derived template retained.

Recommendation and decision conditions

Choose ReplayBench if one teammate is comfortable with SQL and the team can access Tiger Data promptly. The project gives that sponsor an essential role, turns correctness into a visible before-and-after demonstration, and directly addresses the brief's performance emphasis. Use one primary database and avoid an unnecessary AI layer.

Rank	Project and primary target	Decision score	Why it makes the shortlist
1	ReplayBench • Tiger Data	91/100	Strong sponsor dependence, controlled failure/repair demonstration and measurable database work.
2	Handoff Lab • ElevenLabs	86/100	Voice is intrinsic to the practice task; a missing fact becomes a visible ticket error and a repair opportunity.
3	LineCheck • Gemini API	83/100	Useful document extraction supports an exact reconciliation task; the source trail makes model mistakes inspectable.

These are weighted planning judgments, not official judging scores or winning probabilities. Every option is conditional on the eligibility gate. The ranking includes all ten previous ideas and four fresh concepts; no preference bonus is given to prior research.

Assumptions. The user's reply "4" is treated as four teammates. Skills, hardware and remaining build time were not supplied. The baseline assumes ordinary laptops, web/API competence, one SQL-capable person and enough time for the full plan. If SQL capability is absent, ReplayBench falls to 79 under the stated sensitivity test and Handoff Lab becomes the first choice if live voice access works.

Eligibility and scope. The eight challenge briefs are confirmed for this edition by the user. Current ShellHacks submission/reuse details and multi-category entry/award limits remain unresolved. DigitalOcean and GoDaddy are technically compatible additions; no secondary-category eligibility is claimed. The main project should succeed without them.

Evidence classes used throughout: observed historical records and public documentation; user-confirmed edition/team information; proposed features/tests; analyst judgments; and unresolved rules or access. All benchmarks and validation results are future work. No prototype has been built or tested.

Rules and information still needed

Previously verified format: in person at FIU Graham Center, with teams of 1–4 and a 36-hour build window from Friday 25 September 23:00 to Sunday 27 September 11:00, America/New_York. The FAQ admits students and recent graduates within one year from colleges, universities or coding bootcamps. The exact submission cutoff may differ from the end of hacking and must be confirmed. [E26]

Topic	What is established	What remains unresolved
Challenge inventory	Eight cards on the user-confirmed MLH prize page apply to 2026. [M01]	Additional organizer/local sponsor briefs. Sponsor logos are not challenge requirements.
Judging	Tiger Data names innovation, impact and performance. MLH publishes an equal-weight baseline of technology, design, completion and learning for its categories. [M01, M02]	ShellHacks overall rubric, local overrides, sponsor-specific judging procedure and any weights beyond the briefs.
AI tools	The 2027 MLH Member Event Agreement requires allowing generative AI/LLMs; generic rules ask for disclosure. [M02, M03]	ShellHacks disclosure fields and any operational restrictions.
Prior work	Generic MLH rules separate prior ideas from prior project materials and discuss libraries/open source. [M02]	Current event adoption and exact code, assets, design, template and reuse policy. No permission to reuse an earlier app is assumed.
Submission and categories	Cards state the technology/naming targets. [M01]	Official submission URL, cutoff/timezone, required artifacts, video/demo/Q&A duration, opt-ins, entry caps and simultaneous-award limits.

MLH’s standard rules explicitly allow event-specific rules. Their digital-only video requirement is not imported into this in-person event. The ranking weights in this report are our decision framework, not that published judging baseline. No current reuse permission was established, so the build plan does not recommend prebuilt project code. [M02]

Precisely needed to close the gaps. Provide the 2026 guide’s Rules, Judging and Submission sections; the exact current submission URL/form; any extra sponsor cards or opening slides; and the written maximum-category, exclusivity and award-stacking rule. Include code/assets/AI disclosure policy, exact cutoff and demo requirements. Copied text or an export is sufficient; no credentials are needed.

Discovery: official prize page, organizer site/guide and MLH policies searched. Guide content was not available through public reading; prior navigation marks it registered-only. Missing search hits are not proof of absence. Details: research_rules_2026.json.

Challenge matrix technology and judging

The following separates the literal technology target from suggestions and promotion. “Best use” alone is not a published numerical rubric. All four rows below cite the supplied card text and the official page; applicability to 2026 is user-confirmed. [M01]

Challenge	Required technology or action	Stated judging considerations	Optional examples and boundaries
C01 ElevenLabs	Integrate ElevenLabs audio into the project. The brief requests an autonomous audio experience but does not define a minimum architecture.	Best use; no scored rubric in the card. Voice naturalness and expressiveness describe capabilities, not published scoring weights.	AI companions, narrated stories and voice-enabled applications are suggestions. A specific example is not required.
C02 Gemini API	Build an AI application using the Gemini API. Using Gemini only to help write code does not demonstrate product integration.	Best use; no weighted rubric in the card. The invitation to impress is not a measurable threshold.	Advice chatbots, paper summaries and creative generation are optional examples.
C03 Solana	Build a project using Solana. A meaningful on-chain operation is the clearest way to show this; a logo or wallet login alone is weak integration.	Innovation is encouraged; no formal weights or mandatory throughput figure.	Frequent-transaction games, trading/lending/DEX, identity, supply-chain and payment prototypes are suggestions.
C04 Tiger Data	Use Tiger Data to power the project. Select its actual platform/service, rather than assume generic PostgreSQL counts.	The card explicitly emphasizes innovative, impactful and performance-driven use. No weights are supplied.	IoT monitoring, AI analytics and financial prediction are examples. Continuous aggregates and combined relational/time-series data are capabilities, not a requirement to use every feature.

Interpretation. No project is required to use every listed capability. Claims about instant queries, lag-free dashboards, 90%+ compression, near-zero cost, speed of development or naturalness are promotional claims. They are not measured properties of our prototype or guaranteed service entitlements.

Source M01: official MLH ShellHacks prize page and retained user-supplied text. Every category has a common event eligibility/submission gate described in the rules section.

Challenge matrix technology and judging continued

The following separates the literal technology target from suggestions and promotion. “Best use” alone is not a published numerical rubric. All four rows below cite the supplied card text and the official page; applicability to 2026 is user-confirmed. [M01]

Challenge	Required technology or action	Stated judging considerations	Optional examples and boundaries
C05 DigitalOcean	Build using DigitalOcean infrastructure or services. A deployed functional service gives direct evidence of use.	Best use; no card-specific scored rubric or scaling threshold.	Droplets, Managed Databases, App Platform and Gradient AI/GPU/inference are alternatives, not an all-services checklist.
C06 Snowflake API	Integrate a Snowflake API into the application. The card does not require a particular endpoint or a count of models.	Best use; no scored rubric in the card. Speed-of-development claims are promotional.	Custom applications, RAG chatbots and embedded AI are optional examples. Calling many models is not required.
C07 MongoDB Atlas	Build a hack using MongoDB Atlas. A local-only MongoDB database is not evidence of Atlas use.	Best use; no distinct weighted rubric in the card.	No required domain or database feature. Courses, student credits and free-tier access are offers, not mandatory project features.
C08 GoDaddy Registry	Register a domain name with GoDaddy Registry. Eligible extensions and the qualifying registration route are not specified in the card.	Best domain name; no stated rubric, weights or naming restrictions.	No particular naming theme or site type required in the supplied card.

Interpretation. No project is required to use every listed capability. Claims about instant queries, lag-free dashboards, 90%+ compression, near-zero cost, speed of development or naturalness are promotional claims. They are not measured properties of our prototype or guaranteed service entitlements.

Source M01: official MLH ShellHacks prize page and retained user-supplied text. Every category has a common event eligibility/submission gate described in the rules section.

Challenge matrix delivery eligibility and prizes

The cards do not provide a complete submission checklist. Proposed proof is our recommendation, not an additional official requirement. Prize values, item quantities and terms are not inferred where the card leaves them unspecified. [M01]

Challenge	Stated deliverable	Eligibility and access	Prize	Open details
C01 ElevenLabs	A hack using ElevenLabs audio; no separate artifact, video length or API-log format stated.	No card-specific participant restriction stated. Credit-code redemption requires login; credit eligibility is not prize eligibility.	Wireless earbuds; allocation/model unspecified.	Whether simple TTS suffices for the autonomous wording; access/quota/credits; submission process and category stacking.
C02 Gemini API	An application using the API; no distinct video/repository requirements in this card.	No card-specific restriction stated; account/model access must be checked separately.	MLH swag kits; quantities and contents unspecified.	Eligible models/endpoints if ambiguous, quotas, any specific opt-in/evidence and category stacking.
C03 Solana	A Solana-based prototype; network choice, program source and explorer evidence requirements are not specified.	No card-specific restriction stated. Devnet acceptance and any network/token conditions remain unverified.	Ledger Nano S Plus; text describes prizes for each team member.	Devnet/localnet acceptance, whether a custom program is required, RPC/faucet access, entry/award limits.
C04 Tiger Data	Demonstrate how Tiger Data powers the project; no fixed benchmark, compression target or submission format stated.	No card-specific restriction stated. Service/extension qualification and actual account entitlements need confirmation.	Stream Deck Mini; allocation unspecified.	Cloud access/credits and required product variant; event rubric interaction; benchmark conditions; entry/award limits.

ElevenLabs proof to prepare. A live conversational audio loop and visible action/state trace; a cached recording must be labeled.

Gemini API proof to prepare. Actual API-derived output changes the user task; show validated output and a failure/abstention case.

Solana proof to prepare. Signed state transition, rejected invalid transition and explorer evidence on an accepted network; no real funds needed for planning.

Tiger Data proof to prepare. Show a sponsor-dependent time-series operation, a correctness check and a reproducible measured comparison.

Source M01 applies to every row. Product access, credits and prize eligibility are different questions. None of these rows confirms that one project may enter or win several categories.

Challenge matrix delivery eligibility and prizes continued

The cards do not provide a complete submission checklist. Proposed proof is our recommendation, not an additional official requirement. Prize values, item quantities and terms are not inferred where the card leaves them unspecified. [M01]

Challenge	Stated deliverable	Eligibility and access	Prize	Open details
C05 DigitalOcean	A project using its cloud services; no mandatory public uptime, architecture or video format stated.	Account access required; the card directs teams to an MLH Coach for credits, which are not guaranteed by the brief.	Retro wireless mouse; text describes prizes for each teammate.	Credit availability, billing/region limits, required evidence, runtime/deployment constraints, entry/award limits.
C06 Snowflake API	An API-powered project; no mandatory dataset size, model count or deliverable format stated.	The card advertises a special student 120-day trial; account/trial eligibility and enabled models are separate access checks.	Raspberry Pi 4; memory variant/allocation unspecified.	Trial activation, role permissions, model/region access, whether selected endpoint qualifies, submission/stacking.
C07 MongoDB Atlas	A hack using Atlas; no mandated schema, vector search, minimum scale or artifact format stated.	No prize-specific restriction stated. Student credit eligibility differs from standard account/free-tier access.	M5Stack IoT Kit; text describes one for each group member.	Current tier/feature availability, credits, network access, submission evidence and entry/award limits.
C08 GoDaddy Registry	A registered domain; proof, DNS activation and website requirements are not stated.	Code redemption requires login. Exact offer/TLD/registration eligibility must be checked before claiming qualification.	Digital gift card; value and allocation unspecified.	Eligible TLDs and redemption route, registration deadline, renewal terms, naming criteria, category stacking.

DigitalOcean proof to prepare. A working hosted backend and observable deployment/operations evidence; basic hosting alone may offer limited distinction.

Snowflake API proof to prepare. Use an actual Snowflake endpoint for a necessary project function; tie the response to grounded evidence and preserve source identifiers.

MongoDB Atlas proof to prepare. Atlas performs a meaningful persistence, query or consistency function; do not add a duplicate database.

GoDaddy Registry proof to prepare. Eligible registration and a name that communicates the project. Treat this as naming support, not an architecture driver.

Source M01 applies to every row. Product access, credits and prize eligibility are different questions. None of these rows confirms that one project may enter or win several categories.

All ten existing ideas against all eight challenges

This is technical fit, not entry eligibility. S = strong natural role; C = useful only with a bounded adaptation or supporting role; X = weak or forced for the current scope. GoDaddy is a conditional naming addition for every project and receives no architecture bonus. Detailed reasoning for all 80 cells is preserved in research_sponsor_adaptations_2026.json.

Idea	Eleven	Gemini	Solana	Tiger	Digital Ocean	Snowflake	Atlas	Domain
I01 FormTrail	C	C	X	X	C	C	C	C
I02 Contract Canary	X	C	X	C	C	C	C	C
I03 Counterexample Arcade	C	C	X	X	C	C	X	C
I04 WindowWise	C	C	X	X	C	C	C	C
I05 PantryRelay	C	X	X	C	C	X	S	C
I06 LostLink	C	S	X	X	C	C	S	C
I07 LineCheck	X	S	X	C	C	S	C	C
I08 Pulse Path	C	X	X	C	C	X	C	C
I09 StormTable	C	C	X	S	C	C	C	C
I10 BranchBridge	S	C	X	X	C	C	S	C

Reject weak combinations. Solana is forced across the ten original scopes because none establishes a need for decentralized shared state. Do not add a wallet or token solely for a category. Similarly, select Gemini or Snowflake for a given extraction role, and choose one authoritative database.

Scope discipline. A basic deployment may satisfy broad platform-use language while offering little distinctiveness. Tiger Data needs meaningful event history/analytics; a handful of static records is a poor reason to add time-series infrastructure. Pulse Path remains on hold until hardware is confirmed.

The full reasoning and primary vendor sources are recorded in research_sponsor_adaptations_2026.json. All technical fit ratings are analyst judgments; the final ranking compares original and fresh concepts under the same framework.

Adaptations for the original concepts

I01 FormTrail

Keep one form and deterministic completeness rules. Optionally add source-grounded simplification or spoken instructions; retain the original text and accessible controls. Cloud drafts only if cross-device use is tested.

Strongest roles: No strong primary role without changing scope. Reject or defer: Solana, Tiger Data.

I02 Contract Canary

Preserve the replay-and-field-diff core. Add an optional explanation grounded in an explicit trace, or a small persistent regression history. Do not turn one flow into a generic autonomous debugging agent.

Strongest roles: No strong primary role without changing scope. Reject or defer: ElevenLabs, Solana.

I03 Counterexample Arcade

Keep the exhaustive bounded enumerator and animation as the product. Optional narrated steps or grounded hints are small additions; avoid AI-generated counterexamples as the proof.

Strongest roles: No strong primary role without changing scope. Reject or defer: Solana, Tiger Data, MongoDB Atlas.

I04 WindowWise

Retain the six verified service records and deterministic interval solver. An optional natural-language request parser can turn an errand into explicit fields that the user confirms before calculation.

Strongest roles: No strong primary role without changing scope. Reject or defer: Solana, Tiger Data.

I05 PantryRelay

Use Atlas as the sole inventory/reservation store with atomic conditional updates or a transaction, idempotent collection, and explicit expiry processing. Show the last-item race live; omit AI and blockchain.

Strongest roles: MongoDB Atlas. Reject or defer: Gemini API, Solana, Snowflake API.

Each adaptation is proposed work. A sponsor fit rating does not override participant, build-window, reuse, product-access or category-entry rules. The original concept records remain unchanged for comparison.

Adaptations for the original concepts continued

I06 LostLink

Use Gemini image/text embeddings for retrieval and a bounded image-attribute explanation; keep the found-item catalog, metadata and vectors in Atlas. Compare with a keyword/color baseline and demonstrate absent items and similar bags.

Strongest roles: Gemini API, MongoDB Atlas. Reject or defer: Solana, Tiger Data.

I07 LineCheck

Choose one extraction path: Gemini structured document/image extraction, or Snowflake API-backed extraction. Compare confirmed fields with deterministic arithmetic and support source page/crop links only when verified; use two layouts and synthetic documents.

Strongest roles: Gemini API, Snowflake API. Reject or defer: ElevenLabs, Solana.

I08 Pulse Path

Do not select the original hardware build under the software-only assumption. A clearly renamed audio/visual cue-learning simulator can preserve cue-response replay, but it is a changed prototype and cannot substantiate tactile learning.

Strongest roles: No strong primary role without changing scope. Reject or defer: Gemini API, Solana, Snowflake API.

I09 StormTable

Keep every scenario fictional and deterministic. A focused adaptation stores report/decision events and freshness/resource histories in Tiger Data, then replays the same events and explains stale evidence; optional scripted voice briefings can follow.

Strongest roles: Tiger Data. Reject or defer: Solana.

I10 BranchBridge

Keep the authored mystery and deterministic shared state; give roles distinct ElevenLabs voices with captions. Atlas can persist the room and decision trail, with duplicate-action and reconnect checks. Avoid generated plot logic.

Strongest roles: ElevenLabs, MongoDB Atlas. Reject or defer: Solana, Tiger Data.

Each adaptation is proposed work. A sponsor fit rating does not override participant, build-window, reuse, product-access or category-entry rules. The original concept records remain unchanged for comparison.

Fresh concepts from the challenge opportunities

F01 ReplayBench for Tiger Data

Prove whether delayed telemetry changes a dashboard correctly.

Useful sponsor role. Use Tiger Data as the only database: relational device metadata, a time-partitioned event table and one continuous aggregate. Compare event time with arrival time; explicitly refresh affected historical buckets. No AI, blockchain or second database.

Smallest complete concept. Three simulated devices, one integer-valued event schema, a 100,000-row seed stream, late arrivals, unchanged duplicates and a disconnected stream. Controls inject faults; a timeline shows the affected bucket, correction and independent count/sum result. Scale toward one million only after the core works.

Selection risk. Tiger account access and correct refresh semantics are the first gates. Real-time aggregation does not automatically repair old buckets outside its refresh policy. One teammate must be comfortable with SQL.

Historical anchors: P08, P11, P18, P14. Freetear and Pegasus made development friction concrete; Farsense connected telemetry to a visible dashboard; Tango used replay and feedback. These support an inspectable technical workflow, not a causal win claim.

F02 Handoff Lab for ElevenLabs

Practice a support call and repair the exact missing facts in the resulting ticket.

Useful sponsor role. ElevenLabs provides the live caller voice and conversational interaction. The app owns a hidden case fact card, five ticket fields, evidence links and retry state. The human is the trainee. Avoid a second model/provider unless it performs a demonstrably separate necessary task.

Smallest complete concept. Three fictional support scenarios, a live spoken session, editable ticket, transcript evidence and retry of one missed question. The caller can reveal only authored case facts; uncertain transcript extraction needs review.

Selection risk. Voice account/quota, venue microphone/noise and reliable fact grounding. A recording is an honest backup but does not establish a working live loop.

Historical anchors: P05, P14, P27. Speak Free practiced speaking; Tango connected practice and feedback; ARSL made learning interactive. The proposed voice workflow transfers those patterns without requiring VR.

These concepts were generated from the confirmed opportunities and reviewed independently alongside the earlier ideas. Their proposed distinction still needs user testing and is not a claim that no existing product offers related capabilities.

Fresh concepts from the challenge opportunities continued

F03 EvidenceDelta for Snowflake API

Show which source change altered an answer and when the evidence is insufficient.

Useful sponsor role. Use version-filtered Cortex Search retrieval plus actual Cortex REST inference. Stable document/chunk IDs and explicit version metadata separate the two evidence sets; use Snowflake as the sole data/AI stack.

Smallest complete concept. Twenty authored or permission-compatible notices in two preloaded versions; one question, side-by-side cited answers, a changed-fact view and missing/conflicting-evidence state. A text-only corpus avoids OCR.

Selection risk. Snowflake account/role/model access, Cortex Search index readiness and retrieval variability. Confirm API acceptance and costs.

Historical anchors: P24, P18, P32. IntelliSchedule brought relevant information into a task; Pegasus made data queries inspectable; supplementary A11y Tools adapted web information to user needs.

F04 ReceiptRelay for Solana

Record an equipment handoff that two independent organizations can verify.

Useful sponsor role. A Solana program validates asset ID, expected version, current custodian and both authorized signers. One accepted transfer advances state; a conflicting transfer fails. Use a test network only if the challenge permits it.

Smallest complete concept. One fictional asset, two wallets, propose/accept handoff, explorer-linked receipt and a deliberate conflicting-transfer attempt. No token economy or real funds.

Selection risk. Unconfirmed trust need, no known Solana specialist, signing/expiry logic and unverified Devnet eligibility. These are selection gates, not minor stretch features.

Historical anchors: P01, P02. Rent-A-Plug connected a ledger event to a concrete action; Pocket Park connected a resource to a transaction. Historical blockchain use does not establish a present need.

These concepts were generated from the confirmed opportunities and reviewed independently alongside the earlier ideas. Their proposed distinction still needs user testing and is not a claim that no existing product offers related capabilities.

Decision framework and eligibility gate

Eligibility is a prerequisite, not a score. Confirm participant eligibility/attendance, the accepted sponsor product and required use, the event's build/reuse policy and submission opt-in. All twelve ranked candidates are conditional on those checks. ReceiptRelay and Pulse Path have additional hold conditions; neither can be promoted merely by its numeric total.

Criterion	Weight	What a high score means
Challenge fit	20%	Direct response to the actual brief and its stated emphasis.
Useful technical integration	20%	Removing the sponsor removes a central capability; minimal decorative use.
Feasibility	20%	A bounded complete prototype fits the available team and time.
Differentiation	15%	A specific, testable distinction relative to existing products; no global novelty assumption.
Demonstrability	15%	A judge can cause an action and inspect its meaningful consequence.
Dependency safety	10%	Fewer unresolved access, hardware, specialist-skill or service risks. Higher is safer.

Rate each criterion 1–5: 1 is weak, 3 is workable with limits, and 5 is strong under the stated assumptions. Total = sum of weight × rating ÷ 5. Detailed anchors and individual ratings are stored in sponsor_evaluation_2026.json.

Why these weights: 40% rewards a coherent challenge fit and useful sponsor role; 20% protects the ability to finish; 30% covers distinction and demonstration; 10% captures external dependency exposure. These weights are analyst judgments for this team's decision. They are neither learned from the winner dataset nor official MLH judging weights.

Challenge fit asks whether the design answers the brief; integration asks whether the product actually needs the sponsor capability. Feasibility concerns implementation effort; dependency safety concerns external blockers. Some correlation remains. A one-point rating change moves the total by 2–4 points; small total differences should not be treated as precise measurements.

Comparative ranking of fourteen options

Baseline: four software teammates, ordinary laptops, one SQL-capable person and the full 36-hour planning window. Scores do not imply that these assumptions are confirmed. F denotes a fresh concept; I denotes an adapted original. All ranked entries remain eligibility-conditional; equal totals share rank.

Rank	Project	Primary target	Fit	Use	Feas	Diff	Demo	Safe	Total
1	F01 ReplayBench	Tiger Data	5	5	4	4	5	4	91
2	F02 Handoff Lab	ElevenLabs	5	5	4	3	5	3	86
3	I07 LineCheck	Gemini API	5	5	4	2	5	3	83
4	I05 PantryRelay	MongoDB Atlas	5	4	4	2	5	4	81
5	I06 LostLink	Gemini API	5	5	3	2	5	3	79
5	I10 BranchBridge	ElevenLabs	5	4	4	3	4	3	79
7	I03 Counterexample Arcade	DigitalOcean	3	3	5	3	5	5	78
8	F03 EvidenceDelta	Snowflake API	5	5	3	3	4	2	77
9	I02 Contract Canary	DigitalOcean	3	3	4	3	5	4	72
10	I09 StormTable	Tiger Data	4	4	3	3	4	3	71
11	I01 FormTrail	Gemini API	4	3	4	2	4	4	70
12	I04 WindowWise	MongoDB Atlas	3	3	4	2	4	4	66

Fit = challenge fit; Use = useful integration; Feas = feasibility; Diff = differentiation; Safe = dependency safety. Ratings 1–5; total out of 100. The totals are not winning probabilities. Prize monetary value and number of sponsor tools receive no weight.

Held option	Indicative score	Gate that must clear first
F04 ReceiptRelay	71	Unconfirmed cross-organization trust need, test-network acceptance and specialist skill. Do not select on score until those gates clear.
I08 Pulse Path	55	Hardware is not confirmed, and sparse cue logs are weak justification for time-series machinery. A simulator would be a different scoped product.

Preferred project ReplayBench

Prove whether delayed telemetry changes a dashboard correctly.

Challenge satisfaction. Use Tiger Data as the only database: relational device metadata, a time-partitioned event table and one continuous aggregate. Compare event time with arrival time; explicitly refresh affected historical buckets. No AI, blockchain or second database.

Historical evidence. Freetear and Pegasus made development friction concrete; Farsense connected telemetry to a visible dashboard; Tango used replay and feedback. These support an inspectable technical workflow, not a causal win claim. [P08, P11, P18, P14]

Complete MVP. Three simulated devices, one integer-valued event schema, a 100,000-row seed stream, late arrivals, unchanged duplicates and a disconnected stream. Controls inject faults; a timeline shows the affected bucket, correction and independent count/sum result. Scale toward one million only after the core works.

Demonstration. Show a stable total; inject late events; reveal a stale historical aggregate; refresh the complete affected bucket; show agreement with an independent reference calculation. Replay an unchanged duplicate and demonstrate no inflation. Then show measured raw-query versus aggregate-query timings.

Proposed validation. Twelve seeded scenarios covering timely, late, duplicate, missing and combined cases; exact integer count/sum agreement after completed refresh. Run at least thirty comparable queries per path on the same dataset, query semantics and service; report median/p95, environment, warm/cold status and refresh cost separately. These are future tests, not results.

Scope and risk. No production ingestion, arbitrary schemas, anomaly ML, compression promises or exactly-once delivery claim. A stable (event_time, event_id) key handles only unchanged duplicates with unchanged timestamps; account for every partition dimension. Tiger account access and correct refresh semantics are the first gates. Real-time aggregation does not automatically repair old buckets outside its refresh policy. One teammate must be comfortable with SQL.

Primary: Tiger Data. DigitalOcean can host the actual API/UI; a qualifying GoDaddy domain can support the demo. Entry and simultaneous-award permission are unverified, so neither is claimed eligible. Competitor comparison follows. Technology feasibility sources: T01–T03.

Second choice Handoff Lab

Practice a support call and repair the exact missing facts in the resulting ticket.

Challenge satisfaction. ElevenLabs provides the live caller voice and conversational interaction. The app owns a hidden case fact card, five ticket fields, evidence links and retry state. The human is the trainee. Avoid a second model/provider unless it performs a demonstrably separate necessary task.

Historical evidence. Speak Free practiced speaking; Tango connected practice and feedback; ARSL made learning interactive. The proposed voice workflow transfers those patterns without requiring VR. [P05, P14, P27]

Complete MVP. Three fictional support scenarios, a live spoken session, editable ticket, transcript evidence and retry of one missed question. The caller can reveal only authored case facts; uncertain transcript extraction needs review.

Demonstration. Let a judge conduct a short call but miss the location. Show the ticket gap against the case card, retry the relevant moment, and show the corrected handoff. Display captions and a clear live-versus-recorded status.

Proposed validation. Fifteen scripted turns including interruptions and ambiguous answers; compare five ticket fields against human-authored truth. Report field precision/recall, unsupported facts, tool-state consistency and median/p95 turn latency. Five volunteers provide exploratory usability feedback, not efficacy proof.

Scope and risk. One support domain and five fields; no emotion diagnosis, real customers, outbound calls, voice cloning, enterprise integration or broad training curriculum. Voice account/quota, venue microphone/noise and reliable fact grounding. A recording is an honest backup but does not establish a working live loop.

Primary: ElevenLabs. DigitalOcean hosting is technically compatible. GoDaddy is a conditional naming addition. Category-entry and award-stacking eligibility remain unverified. Competitor comparison follows. Technology feasibility sources: V01–V02.

Third choice LineCheck with Gemini API

Turn two small purchasing documents into a reviewable discrepancy report. Gemini performs document extraction; deterministic code compares quantities, prices and totals. This changes the original open-ended OCR plan into one concrete sponsor integration. [G01, G02]

Challenge satisfaction. The project makes actual Gemini API calls and uses the returned fields in the reconciliation workflow, directly matching C02. No requirement forces a chatbot, multiple models or voice. Keep Gemini as the only extraction model.

Historical evidence. CloudPlate linked OCR to records and an action. Supplementary category winner Agent Zero analyzed invoices. Neither establishes validated extraction or fraud-detection accuracy; borrow the inspectable input-to-output workflow. [P13, P33]

Complete MVP. Two supported document layouts, ten synthetic order/invoice pairs, extracted fields with page/quote references, a human review screen and deterministic line arithmetic. Locate a quote in the supported source text before highlighting it; if it cannot be found uniquely, mark it unverified. Do not assume the model supplies reliable text bounding boxes.

Demonstration. Introduce a genuine quantity mismatch and an ambiguous extraction. Correct the extraction; one warning disappears while the real mismatch remains attached to its source. This makes both the useful AI work and its limitations visible.

Proposed validation. Twenty seeded discrepancies across ten pairs plus clean controls. Reserve a subset for evaluation; report extraction field accuracy separately from reconciliation precision/recall and unverified source references. Time five users reviewing a pair versus the same task manually; treat that as exploratory feedback.

Scope and risk. Start with text-readable PDFs or images whose supported layout has verified text anchors. Exclude taxes, currencies, payments, fraud claims and accounting connectors. The main dependency is document extraction with traceable sources. By hour 2, a real API call must extract the required fields on a held-aside sample without inventing evidence.

Primary: Gemini API. DigitalOcean can host the real backend and GoDaddy can provide a qualifying domain, but secondary-entry eligibility is unverified. Atlas is unnecessary unless shared review persistence becomes a tested requirement. Competitor sources: X05–X06.

Distinctive features relative to existing products

The comparison below uses official product pages. It identifies a proposed focus for the prototype, not missing features proven across the entire market. No competitor benchmark or product trial was conducted, and vendor performance claims are not adopted.

ReplayBench compared with Grafana and Monte Carlo

Grafana Metrics Drilldown already explores metrics and related telemetry. Monte Carlo already monitors data freshness, volume and schema. ReplayBench should therefore demonstrate one reproducible fault, an exact count/sum oracle and a repair trace in a compact learning/testing session. Its distinction is that focused exercise, not inventing metrics monitoring or data-quality detection. [X01, X02]

Handoff Lab compared with Yoodli and VirtualSpeech

Both products already support roleplay and feedback; Yoodli supports custom criteria and VirtualSpeech supports custom scenarios. ElevenLabs also has agent-testing features. Handoff Lab should concentrate on the human trainee’s ticket accuracy against an authored fact card and a targeted repair of a missed question. Voice roleplay or automated scoring alone is insufficient differentiation. [X03, X04, V02]

LineCheck compared with DocuClipper and Rossum

DocuClipper advertises purchase-order extraction and invoice matching; Rossum documents configurable invoice-to-PO matching. LineCheck’s prototype emphasis is a small auditable evaluation set, visibly separating extraction errors from true discrepancies and refusing unverified source links. This is a testable presentation/workflow hypothesis, not a claim competitors cannot provide those features. [X05, X06]

Evidence required before a strong originality claim

Use one comparable task and document which distinctive interaction the prototype actually implements. Ask a target user whether that distinction matters. If the difference remains cosmetic, lower its differentiation rating and prioritize a more useful interaction; do not add another sponsor tool to conceal the overlap.

Preferred project responsibilities and early decisions

This plan is proposed work after authorization to implement. Current reuse rules remain unresolved: do not import the existing portal app, prior hackathon code or a prepared project scaffold. At the initial rule check, confirm which SDKs, libraries, templates and assets are permitted and how to attribute them; record AI assistance as required. No prebuilt application code is assumed.

Owner	Primary responsibility	Inspectable handoff
A • Data lead	Tiger schema, continuous aggregate, bounded refresh and benchmark SQL	A delayed bucket repairs to the expected integer count/sum.
B • Service lead	Event ingestion, unchanged-duplicate handling, API, account/hosting setup if allowed	Stable scenario IDs and visible API errors; no hidden mock results.
C • Experience lead	Fault controls, event/arrival timeline, stale/corrected views, accessible UI	A judge can inject a fault and see exactly what changed.
D • Evidence lead	Independent reference calculation, seeded cases, benchmark protocol, product review and submission	Results table, known failures, source/AI attribution and rehearsed demo.

Checkpoint	Go condition	No-go action
First 30 minutes	Verify current rules, primary category opt-in and exact submission cutoff	Keep planning-only status for disputed reuse; choose a compliant from-scratch scope once clarified.
Hour 2	Working accepted Tiger account; event table and continuous aggregate return a result	Switch to Handoff Lab or LineCheck only if that service and the needed skills are available.
Hour 4	One late bucket can be corrected and independently verified	Reduce to one metric/fault. If correctness remains unclear, do not continue scaling or add services.
Hour 7	End-to-end UI → ingestion → aggregate → verified result	Freeze the smallest successful flow; drop secondary targets and optional features.

Do not treat local PostgreSQL, a recording or a cached model response as proof of an unperformed sponsor integration. Account setup and prototype tests have not been executed in this study.

A practical plan for the confirmed hacking window

Full-window baseline: Friday 25 September 23:00 to Sunday 27 September 11:00, America/New_York. Remaining time was not supplied. Rebase to the actual start; if the verified submission cutoff is earlier, move the final freeze and buffer earlier. Breaks and meals belong inside each work block.

Elapsed	Local time	Work and rest allocation	Required outcome
0–3h	Fri 23:00–Sat 02:00	All: confirm rules/skills/access. A/B establish one database flow; C/D define UI, oracle and cases.	Working sponsor connection; project-level go/no-go.
3–7h	Sat 02:00–06:00	All: build one complete late-arrival scenario; brief breaks.	Fault → visible stale state → repair → exact check.
7–13h	Sat 06:00–12:00	A/B rest 6h. C/D build UI and independent tests against agreed fixtures.	Usable fault controls and reviewable ground truth.
13–19h	Sat 12:00–18:00	C/D rest 6h. A/B finish bounded refresh, duplicate handling and API integration.	Core logic passes declared correctness cases.
19–25h	Sat 18:00–Sun 00:00	All: integrate, test 100k rows, measure query paths and failure states.	First complete evidence-backed demo; optional features freeze.
25–28h	Sun 00:00–03:00	A/B rest 3h. C/D evaluate UI and draft demo/submission; no new architecture.	Defect list, draft materials and labeled backup.
28–31h	Sun 03:00–06:00	C/D rest 3h. A/B fix bounded defects and verify deployment if chosen.	Stable demo build and recorded test results.
31–33h	Sun 06:00–08:00	All: rehearse in the actual slot; verify opt-ins, attribution and artifacts.	Submission ready; core feature freeze.
33–36h	Sun 08:00–11:00	Submit before verified cutoff; retain buffer for upload/login/access issues.	Complete accepted entry; no planned last-minute features.

Each teammate has two planned rest blocks totaling 9h. The schedule is a full-window planning scenario, not a claim that all 36h remain. Exact submission timing is an unresolved event rule; do not wait until 11:00 merely because hacking ends then.

Correctness performance and demonstration

ReplayBench should make one defensible claim: for the declared event schema and fault cases, repaired summaries match a separately computed answer. Performance is measured afterward. This is stronger than a large graph whose freshness or semantics are unclear.

Test or measurement	Proposed method	What to report
Correctness	Twelve seeded scenarios; integer count and sum per complete bucket; independent reference calculation.	Agreement/failures after refresh, with scenario IDs.
Duplicate handling	Replay unchanged events with stable event_time and event_id; include every partition dimension in the unique key.	No inflation for supported duplicates; changed timestamps/payloads explicitly outside this identity rule.
Late historical data	Inject events behind the materialization watermark and outside the automatic refresh window.	Stale state, affected complete bucket, completed explicit refresh and corrected result.
Disconnection	Pause a simulated stream while other streams continue; show last arrival age.	Declared simulation and stale indicator; no claim of tested physical sensors.
Comparable query timing	Same data, filters, buckets and results; at least 30 measured runs per path; separate warm/cold conditions.	Median/p95 latency, row count, service size, concurrency and run method.
Lifecycle cost	Measure ingestion and refresh work separately from reads; keep account usage visible.	Refresh duration and available usage/cost data; no universal speed or compression claim.
Usability	Three to five developers attempt one unfamiliar fault explanation after using the UI.	Completion, errors and observations; no broad learning-efficacy claim.

Demo plan. Use a 90-second rehearsal script, then adjust it to the actual judging slot: 20 seconds to show the wrong total; 25 to inject and explain late data; 25 to repair and verify; 20 to show measured tradeoffs and scope. A prerecorded backup must identify what was recorded. This timing is our recommendation, not an official requirement.

Implementation caveats supported by Tiger documentation T01–T03. An unchanged-duplicate key is not a claim of universal exactly-once delivery. Real-time aggregates cover the newest tail; old buckets can still require refresh.

Shorter time alternatives and submission checklist

Situation	Recommended adjustment
SQL skill or Tiger access missing	Prefer Handoff Lab if a real voice/tool loop works; otherwise LineCheck if Gemini extraction and source checking work. Do not keep ReplayBench first by assuming expertise.
About 24 hours available	One replay scenario and 100k rows; omit scaling, extra faults, login, branding and secondary services. Reserve sleep and at least 2h for rehearsal/submission.
About 12–18 hours available	Select the first finalist whose critical service works within 1h. ReplayBench: one metric/fault and a 10k-row correctness demo. Handoff Lab: one case/five fields. LineCheck: one layout/five pairs. Retain a 2h buffer.
Very little time or no working accounts	Use a narrower truthful demonstration. Reassess sponsor eligibility if a fallback removes the required integration; do not present a simulation as completed sponsor use.
Hardware and Solana experience confirmed later	Revisit Pulse Path or ReceiptRelay only if their specific gates and remaining time support a complete core. No automatic promotion.

Prepare once the current submission instructions are available

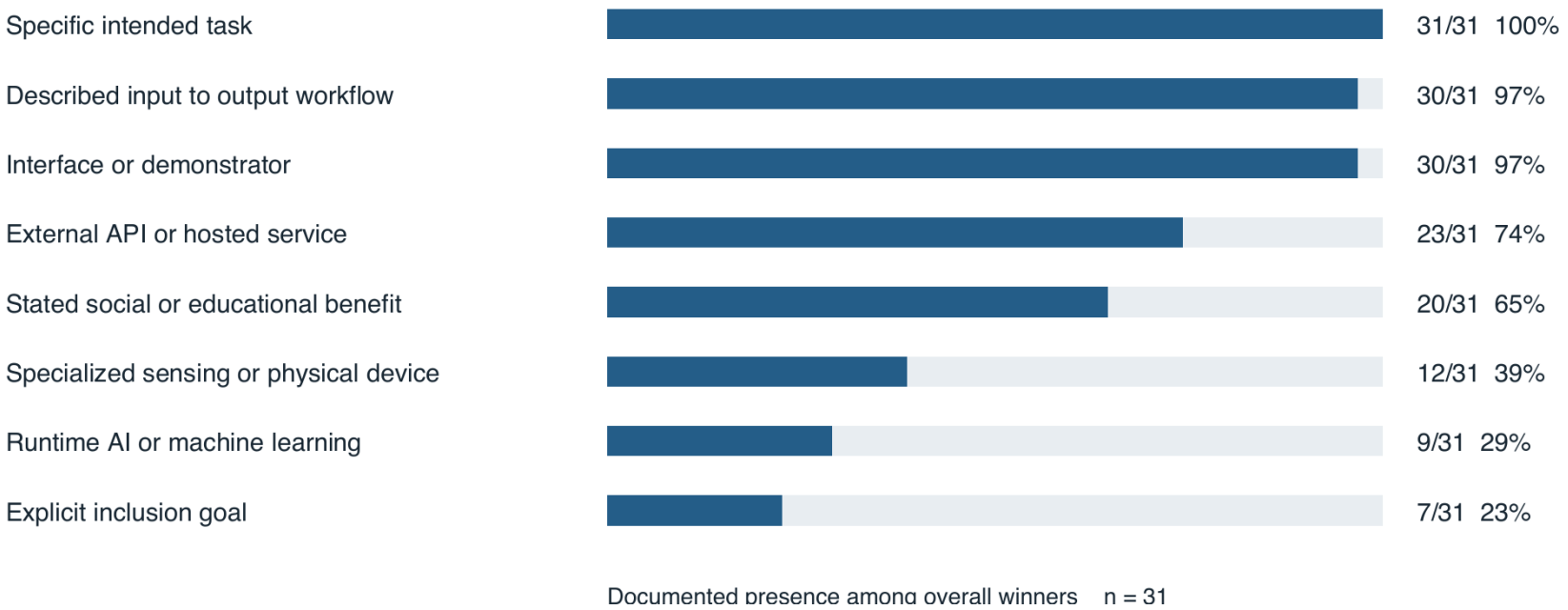
Confirm registration and team details; select the actual primary category; verify whether any secondary entry is allowed; enter the project description and technology attribution; provide exactly the requested repository/demo/video materials; disclose prior materials and AI use according to the current policy; check all links; submit before the confirmed cutoff and retain the receipt.

No secondary category is currently verified. The team should obtain a written answer on maximum category entries, simultaneous awards, exclusivity and any separate sponsor forms/demos. DigitalOcean hosting and a qualifying domain are compatible only if they serve the project and the rules permit entry.

Additional challenge discovery remains open. Evaluate any new organizer brief against this same eligibility gate and scoring framework. A new brief with a better natural fit can change the recommendation; an extra brief does not invalidate this analysis.

How the historical data informs this recommendation

The original evidence remains unchanged: 31 overall podium projects across all nine completed editions, 2017–2025, plus five separately identified 2025 category winners. All 31 overall records state a specific task; 30 describe a workflow and 30 an interface or demonstrator. These are descriptions, not independently verified live performance. [H01]



Documented traits in the unchanged primary cohort, n=31. Categories overlap. “Not established” does not mean absent. Supplementary category winners are excluded from these percentages. Canonical data: dataset.json.

The historical evidence supports concrete tasks, inspectable interactions and a wide variety of technologies. Sponsor fit now provides a stronger selection constraint than choosing a stack from historical frequency. Neither the old data nor the new rubric judgments establish winning odds, easier sponsor categories or a guarantee that a project will win.

Sources and provenance

Challenge wording is retained in challenge_briefs_2026.txt. Edition applicability is user-confirmed; public product documentation supports feasibility analysis only. Product claims and proposed performance targets are not measured results. Reviewed 25–26 September 2026.

M01 MLH ShellHacks prize page

Eight supplied briefs; 2026 applicability user-confirmed.

M02 MLH Standard Hackathon Rules

Published MLH baseline; event-specific adoption/overrides unresolved.

M03 MLH2027 Member Event Agreement

Current member requirements, including generative AI allowance.

E26 Official ShellHacks site and FAQ

Previously verified format; guide/submission details not publicly recovered.

T01 Tiger Data continuous aggregates

Aggregate concepts; configure freshness explicitly.

T02 Tiger Data late-data refresh explanation

Historical buckets and refresh-window behavior.

T03 Tiger Data limitations

Unique indexes include partition dimensions.

V01 ElevenLabs client tools

Conversation-driven UI/tool state.

V02 ElevenLabs agent testing

Existing simulation/testing capabilities.

G01 Gemini document processing

Document understanding; validate extracted evidence.

G02 Gemini structured outputs

Schema structure does not guarantee semantic correctness.

Sources and provenance continued

G03 Gemini embeddings

Multimodal retrieval option; account/model access remains a gate.

A01 MongoDB Vector Search

Catalog retrieval adaptation for LostLink.

A02 MongoDB transactions

Atomic state handling; does not automatically implement application invariants.

D01 DigitalOcean App Platform

Optional functional app/API hosting.

S01 Snowflake Cortex REST API

API access/authentication/model requirements.

S02 Snowflake Cortex Search

Retrieval and index refresh behavior.

S03 Snowflake AI_PARSE_DOCUMENT

Do not assume guaranteed text-line bounding boxes.

B01 Solana transactions

Atomic signed operations.

B02 Solana partial signing

Multiple signers and transaction constraints.

X01 Grafana Metrics Drilldown

Existing metrics exploration and related telemetry.

X02 Monte Carlo data quality

Existing freshness, volume and schema monitoring.

Sources and provenance continued

X03 Yoodli AI roleplays

Existing roleplay and feedback workflows.

X04 VirtualSpeech custom roleplay

Existing custom scenarios and criteria.

X05 DocuClipper purchase-order extraction

Existing extraction and invoice matching.

X06 Rossum two-way PO matching

Existing configurable line matching.

X07 NotLost

Existing photo recognition and lost-property matching.

Study records and retained design

H01 is the unchanged dataset.json and original winner case study. Original ten concepts remain in project_ideas_2026.json. New challenge, evaluation and research JSON files preserve the complete 80-cell matrix, ratings, source links and gaps.

Smartsheet Case Study Report Template supplied the original landscape design, cover illustration and report structure. This revision retains the established typography and table system. Smartsheet does not endorse the research.

Historical project sources supporting the finalists

These project IDs resolve to the unchanged case-study records. Awards and features are participant-reported. P32 and P33 are supplementary category winners. The fuller ten-idea adaptation evidence remains in the original companion data and winner study.

P01 Rent-A-Plug | 2017 Project source

P02 Pocket Park | 2017 Project source

P05 Speak Free | 2018 Project source

P08 Freetear | 2019 Project source

P11 Farsense | 2020 Project source

P13 CloudPlate | 2020 Project source

P14 Tango | 2021 Project source

P18 Pegasus | 2022 Project source

P24 IntelliSchedule | 2023 Project source

P27 ARSL - Augmented Reality Sign Language | 2024 Project source

P32 A11y Tools | 2025 Project source

P33 Agent Zero | 2025 Project source

All new concepts, scores, scopes and validation targets are proposals or judgments. No user accounts, API calls, cloud resources, domain registrations or prototype code were created as part of this study.